

# **Q-Learning Practical Assignment**

## **1. Introduction and Objectives**

Reinforcement learning provides a framework for agents to learn optimal behavior through interaction with an environment and feedback in the form of rewards. In this assignment, Q-learning is applied to a grid-based foraging task, where agents must navigate the environment, avoid obstacles, and collect food resources efficiently.

The study explores both single-agent and multi-agent learning scenarios to understand how learning dynamics change when multiple agents operate simultaneously and potentially compete for shared resources.

### **Objectives**

The main objectives of this work are:

- To implement a Forager Agent using Q-learning in a grid world environment.
- To analyze the learning behavior of a single agent over short (200 episodes) and long (1000 episodes) training horizons.
- To extend the environment to a multi-agent setting with three agents learning concurrently.
- To compare independent Q-tables against a shared Q-table strategy.
- To investigate how learning structure influences performance, competition, and coordination.
- To visualize learning using rewards, food collection, and collision curves as required by the rubric.

Through these objectives, the assignment aims to demonstrate how reinforcement learning can produce adaptive behavior and how different learning strategies affect system-level performance in multi-agent environments.

## **2. Q-Learning Parameters and Justification**

The Q-learning algorithm updates action values according to:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [ r + \gamma \max_{a'} Q(s',a') - Q(s,a) ]$$

The following parameters were used:

- Learning rate ( $\alpha$ ) = 0.15: allows stable yet responsive learning.
- Discount factor ( $\gamma$ ) = 0.95: prioritises long-term rewards.
- Exploration rate ( $\epsilon$ ) = 0.2: maintains continuous exploration while favouring exploitation.

A fixed  $\epsilon$  ensures persistent exploration, which is beneficial in a stochastic environment with multiple agents.

### 3. Single-Agent Learning Behaviour

The single-agent experiments were conducted for 200 and 1000 episodes. The learning performance is evaluated using total reward per episode and the number of food resources collected per episode. Both metrics reflect how effectively the agent learns to navigate the grid and exploit food locations.

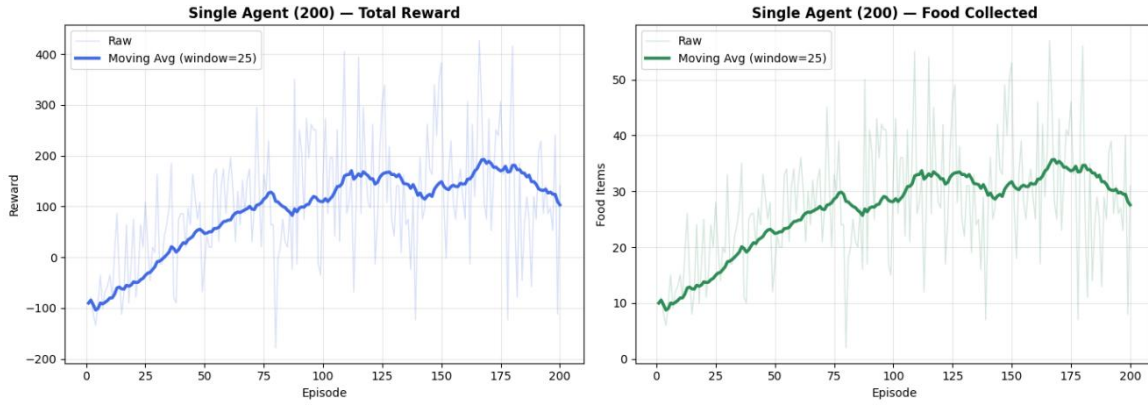


Figure 1: Single Agent (200 Episodes) — Total Reward (left) and Food Collected (right)

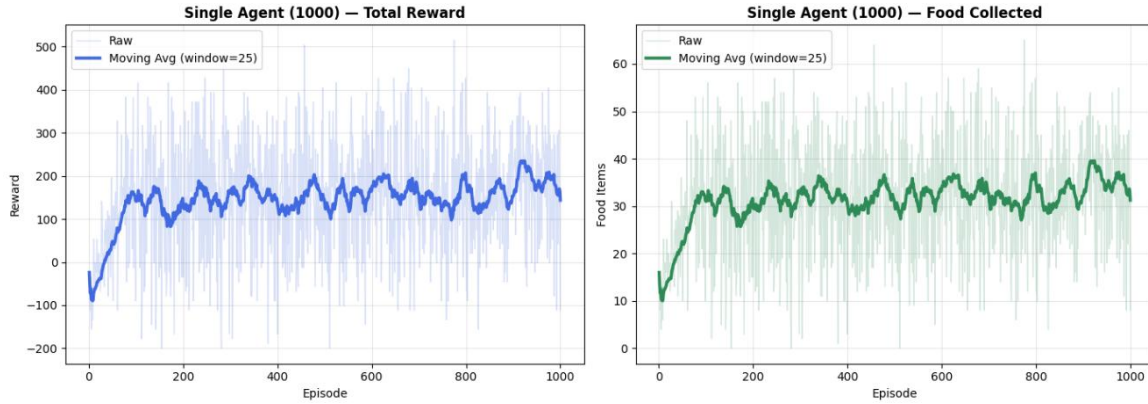


Figure 2: Single Agent (1000 Episodes) — Total Reward (left) and Food Collected (right)

In Figure 1, the curves show an initial low performance followed by a steady increase, indicating that the agent is learning useful state-action values. In Figure 2, the curves stabilize at higher values, demonstrating convergence towards a near-optimal policy. The remaining fluctuations are expected due to  $\epsilon$ -greedy exploration, which ensures continued sampling of alternative actions.

### 4. Multi-Agent Learning Behaviour

The environment was extended to include three forager agents learning simultaneously. Two strategies were evaluated: independent Q-tables, where each agent learns its own policy, and a shared Q-table, where all agents update and exploit a common policy.

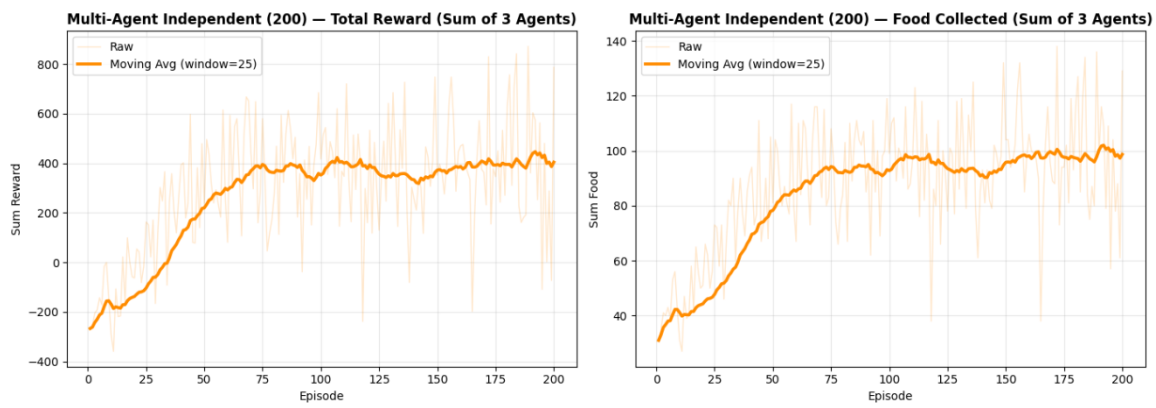


Figure 3: Multi-Agent Independent (1000 Episodes) — Reward (left) and Food Collected (right)

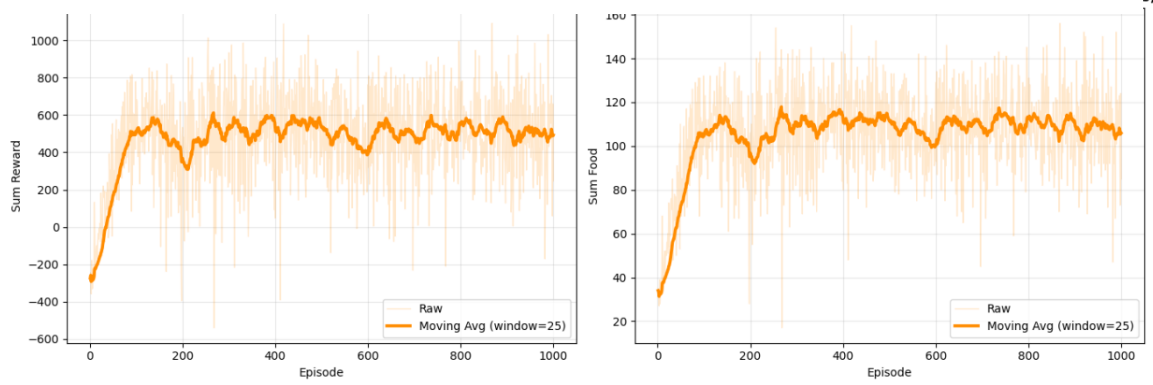


Figure 4: Multi-Agent Independent (1000 Episodes) — Reward (left) and Food Collected (right)

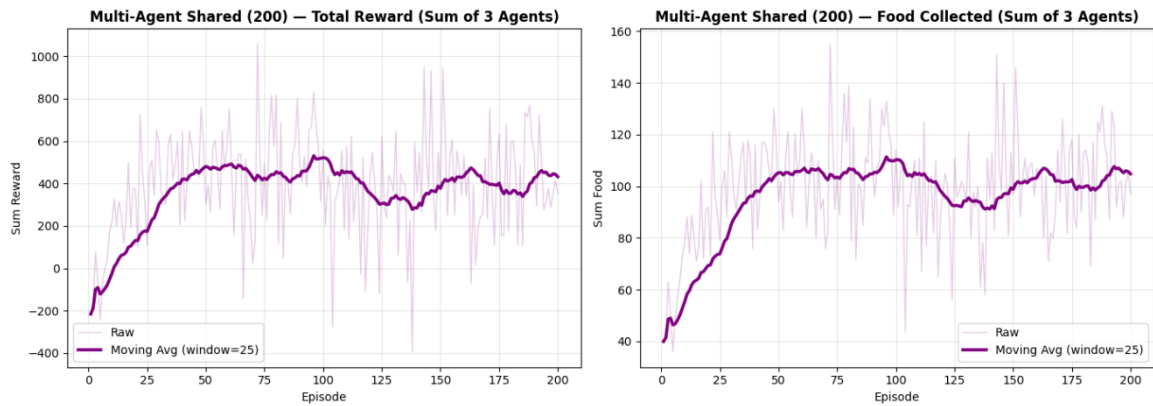


Figure 5: Multi-Agent Shared (200 Episodes) — Reward (left) and Food Collected (right)

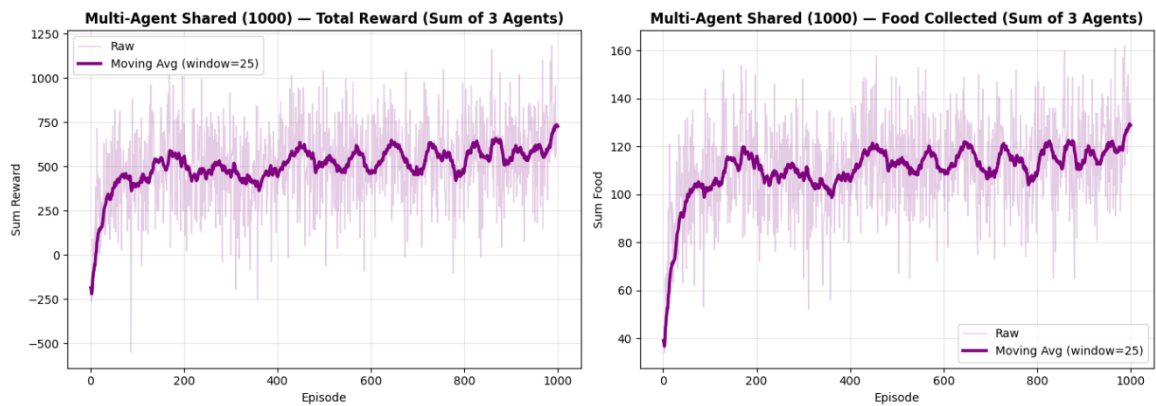


Figure 6: Multi-Agent Shared (1000 Episodes) — Reward (left) and Food Collected (right)

As shown in Figures 3 and 4, where agents learn with independent Q-tables, the total reward and food collected increase gradually over time. The curves indicate steady learning, but with slower convergence. This behavior is expected because each agent treats the other agents as part of a constantly changing environment. As all agents update their policies simultaneously, the learning problem becomes non-stationary, making it harder for any single agent to accurately estimate long-term rewards.

In contrast, Figures 5 and 6 demonstrate that the shared Q-table setup achieves faster early improvement in both reward and food collection. Since all agents update and exploit a common Q-table, experiences gathered by one agent immediately benefit the others. This leads to quicker propagation of useful knowledge and faster policy improvement. The steeper initial rise of the curves reflects improved sample efficiency. However, because all agents follow similar policies, they are more likely to target the same high-reward states, which can increase competition for food resources.

## 5. Independent vs Shared Q-Table Comparison

To directly compare both learning strategies, reward and food collection curves were plotted together. This allows observation of convergence speed and final performance under both approaches.

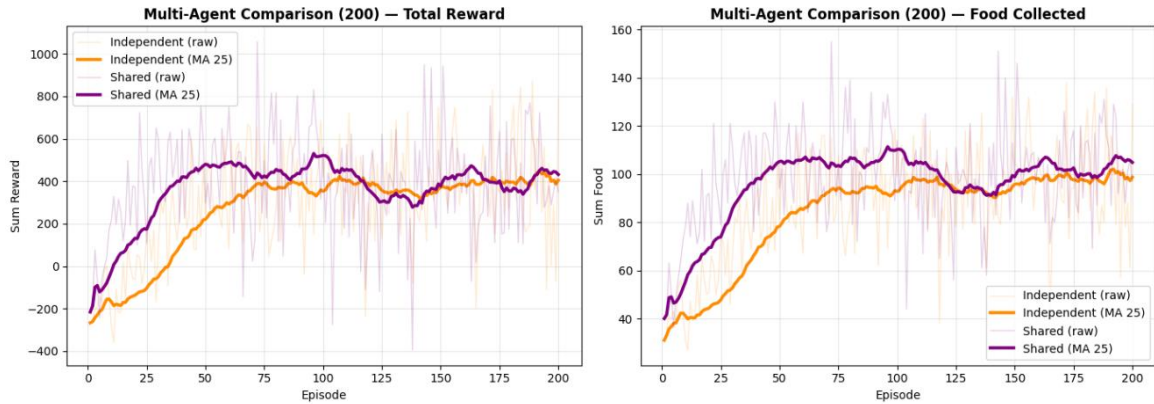


Figure 7: Comparison (200 Episodes) — Independent vs Shared Reward (left) and Food (right)



Figure 8: Comparison (1000 Episodes) — Independent vs Shared Reward (left) and Food (right)

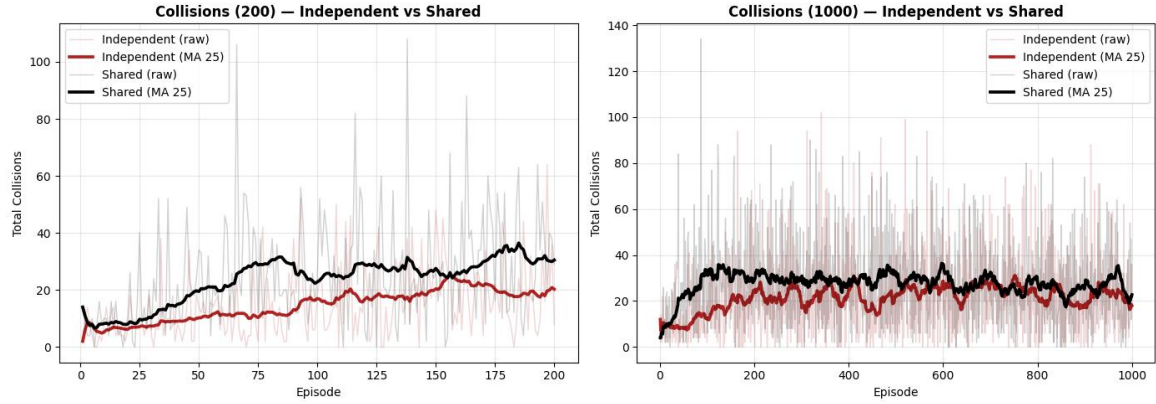


Figure 9: Collisions Comparison — Independent (left) vs Shared (right)

The direct comparisons in Figures 7 and 8 show that the shared Q-table consistently achieves higher performance than the independent approach for both total reward and food collected. This confirms that sharing experiences accelerates convergence and allows the system to learn effective foraging strategies more quickly. The independent approach improves more slowly but shows smoother, more diverse learning behavior, reflecting agents developing different policies.

The collision trends in Figure 9 provide further insight into agent interactions. In the shared setting, collisions are initially higher, indicating that agents often move toward the same cells as they follow similar policies. This reflects increased competition arising from policy alignment. Over time, however, collision rates decrease in both approaches. This suggests that agents gradually adapt their movement patterns and spread out across the grid, leading to emergent coordination and implicit spatial separation, even without explicit communication.

## **6. Conclusion and Future Improvements**

This study demonstrates that Q-learning can effectively solve a grid-based foraging task in both single-agent and multi-agent scenarios. Single-agent results confirm stable convergence and improved food collection over time. In multi-agent settings, shared Q-learning accelerates learning but may increase early competition, while independent learning provides more diverse behaviours.

Potential improvements include:

- Using a decaying  $\epsilon$  to gradually shift from exploration to exploitation.
- Increasing state representation with local obstacle or agent awareness.
- Extending to larger grids or dynamic environments.
- Evaluating alternative multi-agent RL methods such as policy gradients or centralized critics.

Overall, the experiments highlight key challenges and opportunities in cooperative and competitive multi-agent reinforcement learning.